



ESPE

UNIVERSIDAD DE LAS FUERZAS ARMADAS

INNOVACIÓN PARA LA EXCELENCIA



UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE
COMPUTER SCIENCE DEPARTMENT
ADVANCED WEB DEVELOPMENT - 27819
HOMEWORK # - SLIDES NOTES

Slides: Introduction to Fast Api

FastApi building APIs quickly and efficiently

SQLAlchemy can interact better and professional with Databases

Both are important for backends.

Slides: Foundations of User Interface Design

1st Law: A computer shall not harm your work

2nd Law: A computer shall not waste your time

3rd Law: An interface is humane if its responsive to human needs

4th Law: The user should set the pace of interaction

• Methodologies

• ~~Methodologies~~

• Interface is in according understanding of the users, their needs, limitations, user intentions.

Slides: Cognetics and Locus of Attention

We have some points of attention for users.

• Automation of behaviors

• Reduce number of click

• Use terms that familiar to user. User should understand

We have design GUI to accommodate limitations of users.

Slides: Getting started with UI design

UI design focuses on helping users achieve their goals.

We should understand user motives, intentions, behaviours and goals.

An important point is users have different learning levels

- Experts, Occasional, Beginners.

Main UI design methodologies:

- Human-Centered design
- User-Centered design
- Usage-Centered design

User-CD uses people, user profiles and scenarios.

Usage-CD focus on tasks, use cases, and user roles.

Interface should adapt to user motivation, skill level, and behavior patterns.

UI design is challenging due to increasing complexity, devices, and expectations.

Slides: Information Organization

UI content organization starts with understanding requirements

Scenarios should be divided into logical conversation units

Loosely coupled

Highly cohesive

Reusable

Interface metaphors help users understand the system

using familiar concepts.

Interface idioms are recognizable UI styles with consistent visual and actions.

Design patterns provide proven solutions to recurring UI problems

Patterns help decide what to show, when to show it and why

Pattern	what	when	why
	3 elements on the main page	long list of items	Searching and Browsing go hand in hand.

Slides: Getting Started with User Research

User research helps understand real user goals.

Knowing users requires research, not assumptions.

Asking why? repeatedly helps uncover real needs.

Interfaces must adapt to user motivation, skill and involvement.

Designers must balance beginner and expert users.

Increasing complexity devices and expectations make UI design harder.

Interface metaphors and idioms help users understand and interact with systems.

Slides: Page Layout

Page layout guides user attention to convey meaning, sequence.

Layout follows rules and rationale.

Five elements

- Visual hierarchy
- Visual flow
- Grouping and alignment
- Dynamic displays
- Integration of elements

Clear focal points guide user attention.

Slide: Slides - Navigation

Singpost help users stay oriented

Common singposts included

- Window titles
- Logos
- Branding
- Highlighted tabs
- Selection indicators

Wayfinding is how users move toward their goals

Designers must support wayfinding through wayshowing

Navigation models define system structure

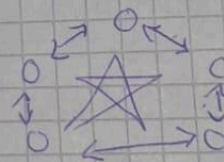
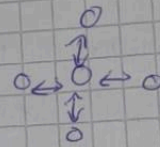
- Hub and Spoke
- Pyramid
- Multi-level
- Flat navigation
- Stepwise
- Model panels

Layout + navigation patterns include

- Flat menus
- Sitemap footers
- Sing-in tools

Navigation design must consider

- Audience
- User goals
- Cognitive load
- Navigation cost
- Workflow efficiency



Slide: Actions and Commands

Make important actions

- Visible
- Clearly labeled
- Easy to find

Support action sequences, automation

Action/command mechanisms can be

- Visible (buttons, menus, toolbar, links)
- Invisible (Keyboard shortcuts)

Basic UI guidelines

- Use visual affordances
- Change cursor on hover
- Provide tooltips

Action and command patterns include

- Button groups
- Action panels
- Previews
- Progress indicators
- Cancelability

Advance Patterns

- Multi-level undo
- Command history
- Macros
- Intelli-Sense
- Multiple gestures for one action

Slide: Showing Complex Data

Showing complex data focuses on visualization data to convey knowledge

Information graphics use

- Charts
- Tables
- Trees
- Maps

Key questions include:

- How data is organized
- How data relates to real-world concepts
- How users can explore and rearrange data

Developing information graphics require

- User research
- System and data analysis

Common data organization models

- Linear
- Network
- Tabular
- Geographic
- Hierarchical

Shape and appearance convey meaning

Precisive variable include

- Color, size, position, oriented, texture, shape

Filtering and querying should be

- Interactive
- Iterative
- Contextual

Labels, legends, axes and datatips help give data meaning.